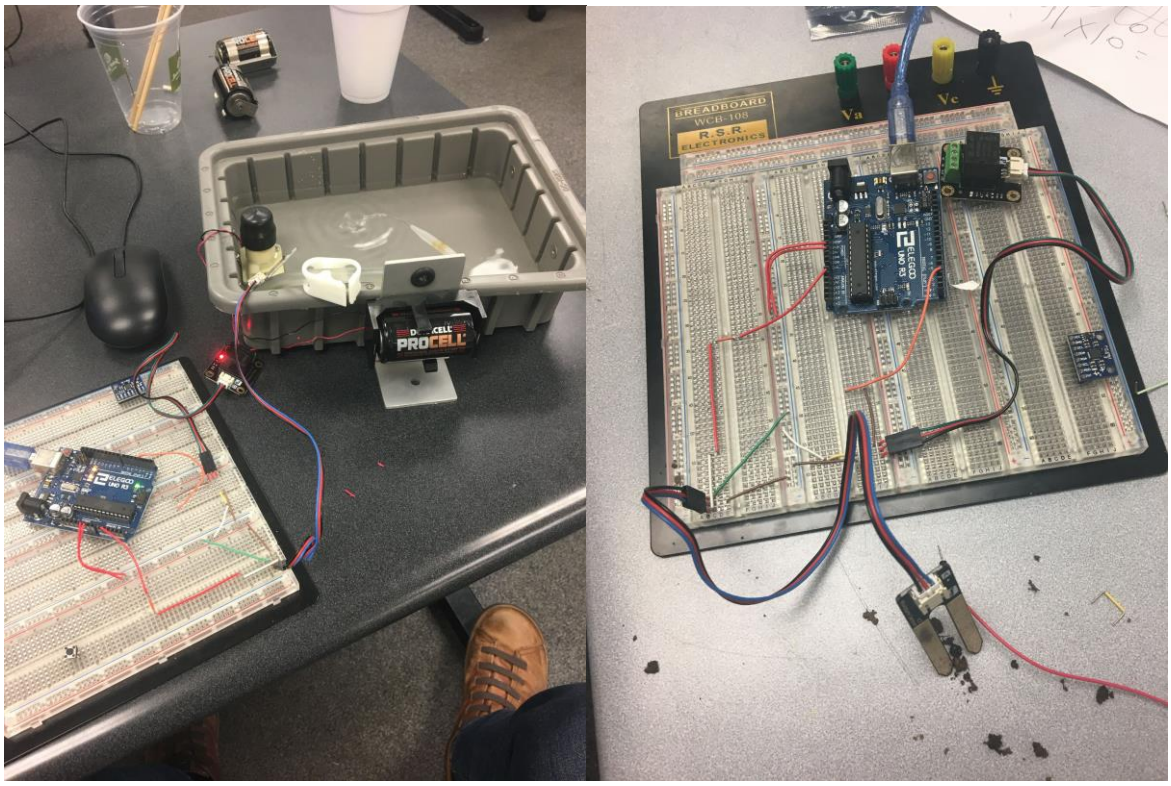
Automated Greenhouse Operations

Introduction:

STLCC Engineering club required an exhibit for earth-day 2026, and so, while everyone else was working on other projects, I was set on building the exhibit. The general concept of the project was to utilize a Soil moisture sensor combined with an Arduino uno and a pump to keep a constant level of moisture in the soil of a singular plant.

Design:

One standard 5v switching JOC-3FF relay was affixed between a 9v battery and the motor that ran the pump; to supply the 5v activation threshold of the relay an Arduino is used with a USB battery pack, and with the Soil moisture sensor plugged into one of the analog read pins. (pictured below)



Then a frame for the greenhouse was made from 1/8thin wide 1ft long wooden dowel rods, it was primarily formed from two 1ft cubes and a pair of equilateral wedges, covered in a folded plastic sheet. (pictured below)



Additionally a temperature and humidity sensor was added to determine the state of the inside of the greenhouse.

Conclusion:

The greenhouse was demonstrated twice by the engineering club, once for earth day and again for Sinco Demio (in a deconstructed form), unfortunately I was unable to be present for either so for the first showcase its battery was accidently disconnected and it didn't run, and for the second demonstration it rained and the system drowned; however, bench tests showed promise, and it was functional after the battery was reconnected the first time.